Mathematics Learning Experience Analysis and Redesign

Grade 7 Mathematics — Fractions and Decimals

Basis/guide: DepEd Mathematics 7 Quarter 1 Module 5, Expressing Rational Numbers from Fraction Form to Decimal Form and Vice-Versa. The module states that learners should express rational numbers from fraction form to decimal form and from decimal form to fraction form.

Selected existing experience: The “What’s More” activity, particularly the exercises asking learners to convert fractions to decimals and decimals to fractions. The module asks learners to round fraction-to-decimal answers to the thousandths place and to express decimals as fractions in lowest terms or as mixed numbers.

Part A. Context and Learning Need

Item	Answer
Grade level and mathematics topic	Grade 7 Mathematics. Topic: expressing rational numbers from fraction form to decimal form and vice versa.
Target competency or learning outcome	Learners should be able to express rational numbers from fraction form to decimal form and from decimal form to fraction form. For fraction-to-decimal work, learners also practice rounding answers when required.
Learner prior knowledge	Learners need knowledge of fractions and decimals, operations on integers, and greatest common factor (GCF). The module identifies the previous lessons on operations on integers and GCF as useful preparation for this lesson.
Likely misconception or difficulty	Learners may confuse the numerator and denominator, place the decimal point incorrectly, or fail to simplify a decimal-to-fraction answer. They may also have difficulty converting fractions whose denominators cannot easily be changed to 10, 100, or 1000 and may need an approximate decimal.
Available devices and connectivity	A typical classroom may have a teacher's phone/laptop and projector, while learners may have limited or no personal devices. The redesigned activity therefore works

	without internet and uses technology only as an optional support.
Selected digital or non-digital resource	Primary resource: printed fraction/decimal cards, place-value chart, paper strips, pencil and worksheet. Optional digital resource: calculator or spreadsheet/calculator app for checking decimal approximations. The DepEd module itself remains the main content guide.

Part B. Analysis of the Existing Experience

Analysis Lens	Evidence from the Existing Activity	Judgment and Implication
Learner mathematical actions	Learners convert fractions to decimals and decimals to fractions by following procedures and completing exercises. The module explicitly provides steps for both directions.	The activity gives practice and procedural fluency, but the redesign should add opportunities for learners to explain why the procedures work.
Constructivist or experiential opportunities	The existing module presents worked examples and then provides exercises. Examples include 3/4 → 0.75 and 0.75 → 3/4.	Useful for guided learning, but learners can be more active by predicting, manipulating representations, and discovering equivalent forms before formalizing the rule.
Collaboration and discourse	The exercises are mainly individually presented in the module.	Add pair/group discussion so learners compare methods, justify answers, and explain errors.
Cognitive-load risks	The module contains several procedural steps, including multiplying numerator and denominator and placing the decimal correctly.	Chunk the procedure into short steps and keep a visible worked example/place-value guide. This reduces unnecessary load without reducing the mathematical task.
Multimedia design	The source is a self-learning module with written explanations, examples, exercises, and answer keys.	A redesign can use one simple visual demonstration rather than many media elements. Visuals should directly support the mathematical idea.
Representation connections	The module connects	Strengthen the connection

	fraction notation and decimal notation, including tenths, hundredths, and thousandths.	among fraction symbols, place-value/decimal representations, and verbal explanations.
Feedback and assessment	The module includes pre-test, activities, post-test/self-check, and answer keys.	Keep short formative checks during the activity so misconceptions are addressed before the final assessment.
Accessibility and inclusion	The module states that its language recognizes diverse vocabulary levels and that learners may consult the teacher/facilitator when they have difficulty.	Use clear instructions, large-print cards when needed, peer support, and a non-digital option so participation does not depend on device access.
Mathematical accuracy and reasoning	The module gives explicit conversion procedures and worked examples. It also requires rounding in some exercises.	The mathematics is appropriate for the target skill. The redesign should preserve the exact conversion and simplification requirements while increasing reasoning.
Feasibility and contingency	The module was designed for use in different learning situations and includes home-based learning guidance.	Use inexpensive paper materials and make the digital checking component optional. If devices or connectivity fail, the whole activity can still proceed.

Part C. Redesigned Learning Experience

Redesigned activity title: "Fraction–Decimal Match and Explain"

Learners work in pairs. They first predict matches, then use fraction/decimal representations to justify the match, and finally solve new conversion problems. The activity preserves the module's target skill while adding exploration, discussion, and formative feedback.

Lesson Phase	Teacher Move	Learner Mathematical Action	Technology / Resource Function	Evidence Collected
Elicit prior thinking	Review fractions, decimals, place value, and GCF. Show 3/4 and 0.75 and ask learners to think about how they may represent the	Recall prior knowledge and identify what the numerator, denominator, and decimal places represent.	Paper place-value chart; optional board/slide.	Oral responses and quick diagnostic answers.

	same quantity.			
Prediction or initial representation	Give pairs fraction cards and decimal cards. Ask them to predict which cards belong together before calculating.	Match equivalent fraction and decimal forms and explain their initial reasoning.	Printed cards; no internet/device required.	Initial matches and written justification.
Guided exploration / experience	Guide learners through examples using the module procedure: make the denominator 10, 100, 1000 when possible; multiply numerator and denominator by the same number; then place the decimal correctly.	Convert selected fractions, record the equivalent decimal, and check the place value.	Place-value chart and worksheet; optional calculator for checking.	Completed conversion table and teacher observation.
Discussion and comparison	Ask pairs to compare methods and discuss why 3/4 becomes 75/100 and therefore 0.75.	Explain and defend a solution; identify and correct an incorrect solution.	Board/document camera or paper work sample.	Student explanations and corrected examples.
Formalization / abstraction	Summarize the two conversion procedures. For decimal-to-fraction, emphasize writing the decimal over 1, multiplying by 10/100/1000 according to the number of decimal places, then simplifying.	Write the general steps in their own words and apply them to an example such as 0.75 = 3/4.	Module/teacher notes as reference.	Short guided practice.
Application / transfer	Give new problems and one simple context problem, such as a distance represented by 4/5 mile, requiring fraction-	Solve independently, show the procedure, and explain the meaning of the answer.	Worksheet; optional calculator only for checking.	Independent solutions and explanation.

	to-decimal conversion.			
Assessment and feedback	Use a 5-item exit ticket: 2 fraction-to-decimal, 2 decimal-to-fraction, and 1 explanation/error-analysis item. Give immediate feedback on common errors.	Complete the exit ticket and revise at least one incorrect answer.	Printed exit ticket; answer key/teacher checking.	Exit-ticket scores plus corrected work.

Part D. Design Justification

1. Primary learning theory/theories

The redesign is primarily informed by constructivist and experiential learning. Learners actively construct understanding by predicting, matching, manipulating representations, discussing solutions, and then formalizing the procedure. It also uses social interaction through pair discussion.

2. Reducing unnecessary cognitive load

The redesign keeps the mathematical challenge but removes avoidable confusion by presenting the procedure in small steps, using a place-value chart, modeling one example, and keeping directions visible. Learners are not required to handle unnecessary technology or complicated instructions.

3. Connecting at least three representations

The lesson connects (1) fraction notation, (2) decimal/place-value representation, and (3) verbal/written explanation. For example, learners connect 3/4, 75/100, 0.75, and the explanation that these represent the same quantity.

4. Technology contribution

A calculator or spreadsheet/calculator app can make checking decimal approximations faster and can help learners notice patterns. Technology is used as a checking/support tool rather than replacing the mathematical reasoning.

5. One risk or limitation and management

A likely limitation is unequal access to devices or connectivity. The activity is therefore designed to work completely offline with printed cards, paper, and a place-value chart. Digital checking is optional.

6. Accessibility and low-resource/offline alternative

Use large, readable cards; short directions; teacher modeling; pair support; and multiple ways to show understanding through matching, writing, speaking, or drawing a representation. All essential tasks can be completed with paper-based materials.

7. Transferable mathematical understanding

The final assessment includes both directions of conversion and an explanation/error-analysis item. This checks whether learners can apply the process to unfamiliar numbers and explain their reasoning, rather than only repeat the exact examples from the module.

Suggested Materials and Teacher Guide

Material	Purpose
Fraction cards	Represent fractions that learners will match and convert.
Decimal cards	Provide equivalent decimal representations for matching.
Place-value chart	Support correct placement of tenths, hundredths, and thousandths.
Worksheet/paper	Record procedures, explanations, and answers.
Calculator — optional	Check decimal approximations after learners solve.

Teacher note: The original module explicitly teaches fraction-to-decimal conversion by changing the denominator to 10, 100, 1000, or another power of ten when possible, then placing the decimal according to the number of zeros. For decimal-to-fraction conversion, it teaches writing the decimal over 1, multiplying by the appropriate power of 10, and simplifying. The redesigned experience keeps these procedures as the mathematical core.

Reference Used

Department of Education. Mathematics 7, Quarter 1 – Module 5: Expressing Rational Numbers from Fraction Form to Decimal Form and Vice-Versa. First Edition, 2020. The module identifies the lesson outcomes and provides the conversion procedures, examples, activities, and assessments used as the basis of this redesign.